Computer Controlled Hand-held Game Station

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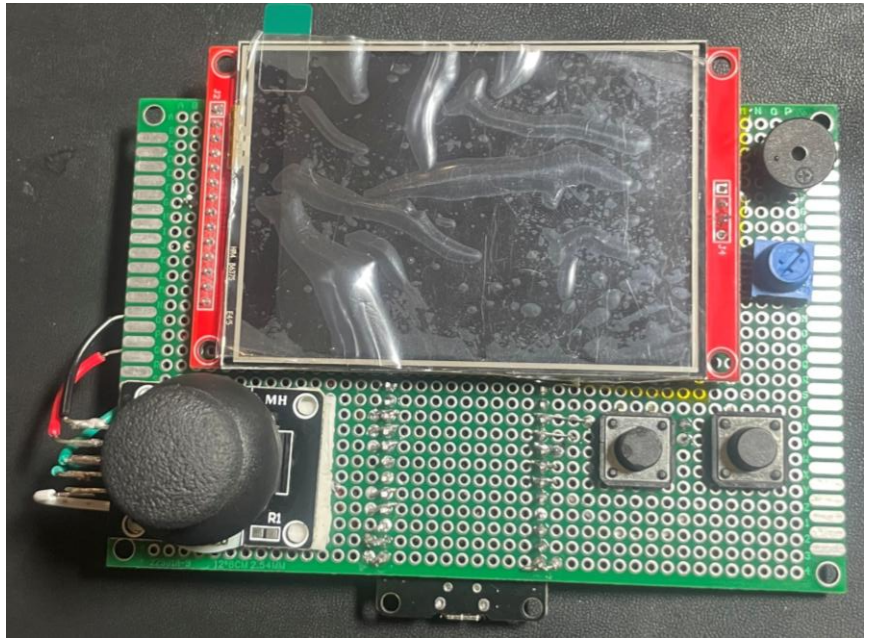
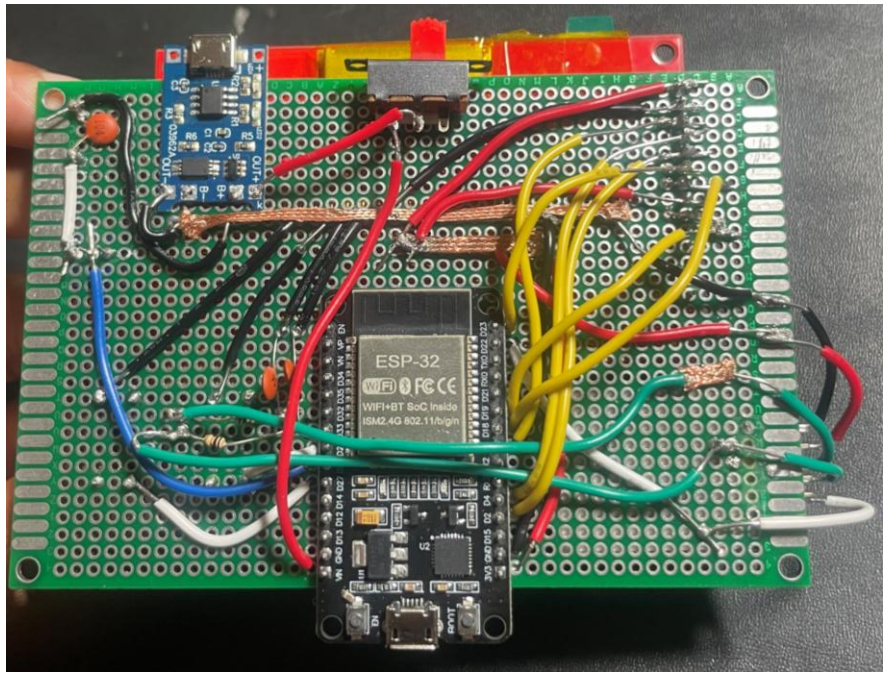
Summary

ESP32-Based custom handheld gaming device integrating a SPI TFT display, input controls, and battery-powered hardware into a compact form factor. Designed circuit layout to incorporate analog and digital inputs while mitigating interference. Implemented firmware for SPI peripherals and GPIO-driven buttons/joystick, including debouncing, menu navigation, and basic game loops under tight timing and memory constraints. Developed the power subsystem with Li-ion battery, charging, and regulation, and validated the design through iterative prototyping, oscilloscope-based signal debugging, and performance testing. Documented hardware schematics, PCB layout decisions, and firmware architecture.

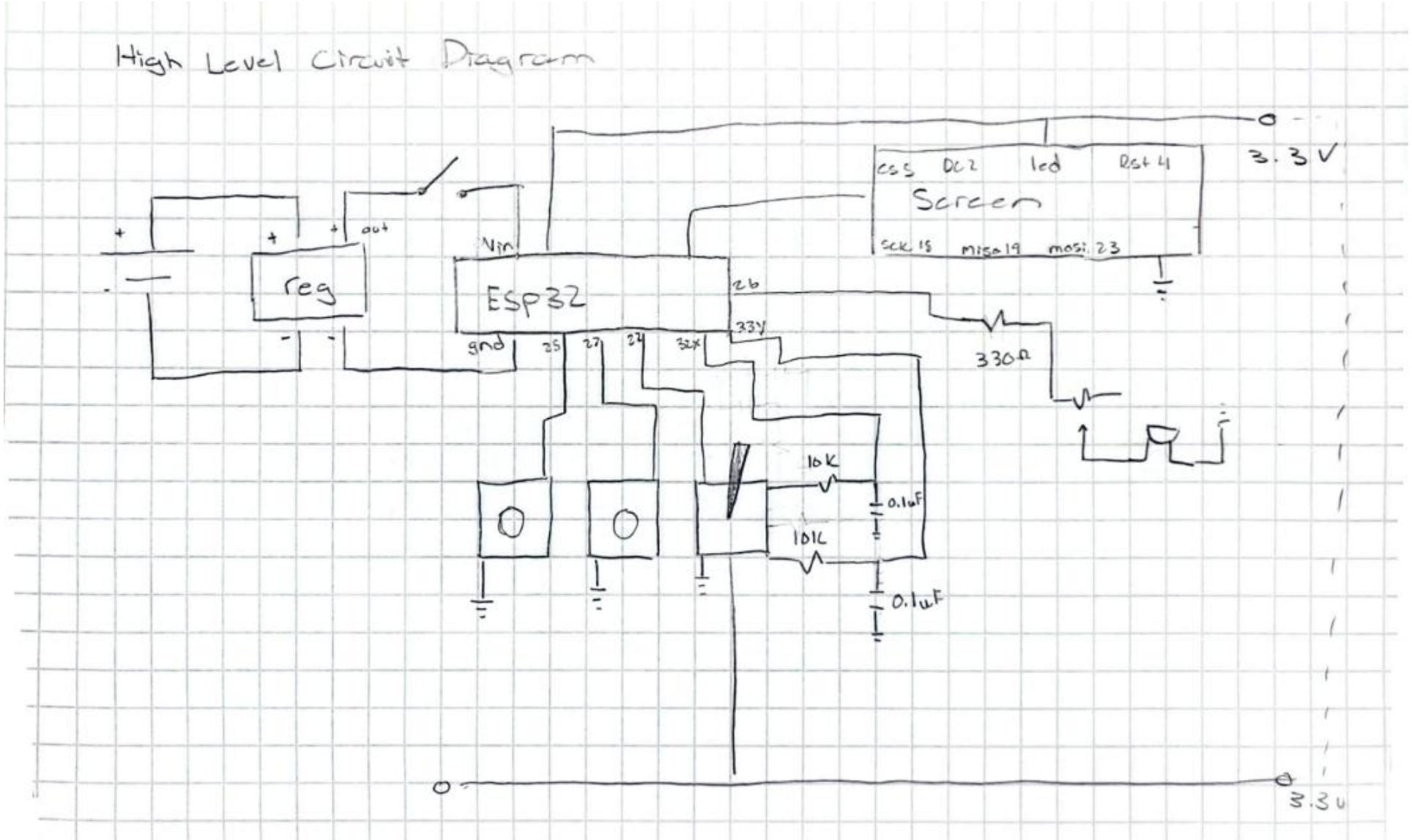




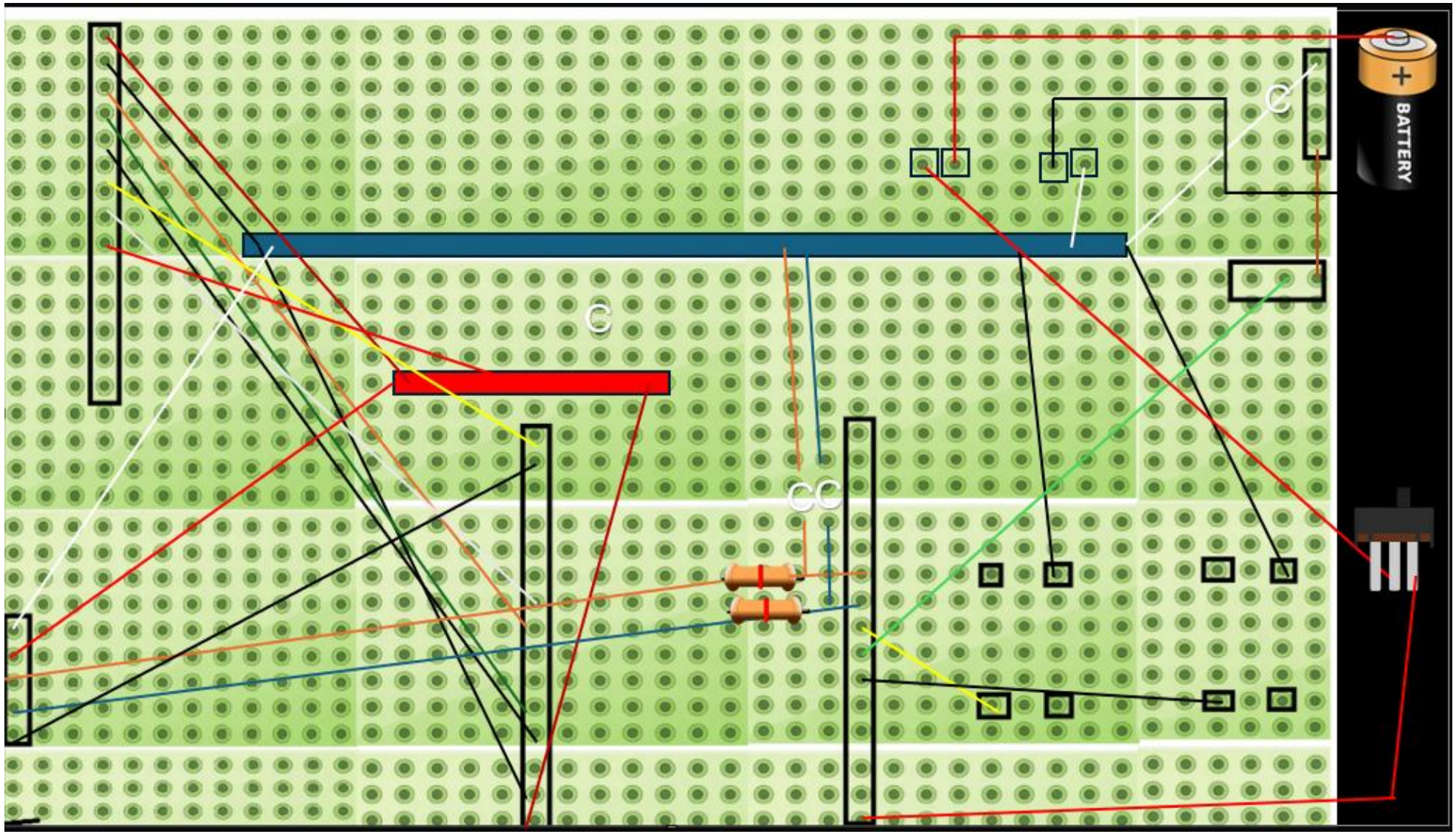
Bill of Materials

- Haldzemo 3.2" TFT LCD Module 320x240 SPI Serial ILI9341 Resistive Touch Screen with Touch Pen Compatible with Arduino STM32
- Joystick Module for Arduino, ESP32, ESP8266, Raspberry Pi – Analog Control Stick for Robotics and Projects, 2-Pack
- DEVMO 2pcs TP4056 Charging Module 5V Micro USB 1A 18650 Lithium Charging Board with Protection Charger Module with Dual Protection Function
- LampVPath (Pack of 2) PCB Prototype Board, Double Sided PCB Proto Board for DIY Solder Breadboard, 8cm X 12cm(3.150" X 4.724") Universal Printed Circuit Board
- ESP-WROOM-32 ESP32 ESP-32S Development Board 2.4GHz Dual-Mode WiFi + Bluetooth Dual Cores Microcontroller Processor Integrated with Antenna RF AMP Filter AP STA Compatible with Arduino IDE (3PCS)
- Lithium 3.7 V, 800 mAh, rechargeable battery

Circuit Diagram



Perf Board Diagram



Engineering Considerations

- Low pass filtering for analog Joysticks
- Charging unit placement away from analog components
- Decoupling Capacitors across rails and buzzer
- Ground rail resistance was too high and cause inconsistent battery connection, fixed by applying copper braid over rail
- Joystick analog signals not getting through, fixed by replacing the joystick then using wiring to get around the orientation issues
- Potential issue that may happen, screen may struggle to get power from high resistance power rail so may need to use braid technique again
- No space for SD pins so will have to solder them on the screen before placing into perf board
- Heat sink on charging unit to reduce heat